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Novel and versatile room temperature ionic liquids for energy storage†

Mega Kar,^a Oscar Tutusaus,^b Douglas R. MacFarlane^a and
Rana Mohtadi^{*b}

Due to their high glass transition temperatures, ionic liquids based on closo-boron clusters have been long discounted from consideration in energy storage applications. Here, we report a strategy that resulted in a novel family of closo-boron-cluster based room temperature ionic liquids (RTILs). Their very low glass transition temperatures, high cathodic and anodic stabilities and compatibility with Li and Mg metals make them excellent candidates as electrolytes for rechargeable batteries and hybrid devices.

Over the past few decades, certain families of ionic liquids have emerged as potentially safer electrolytes in the field of energy storage, due to their low-flammability, negligible vapour pressure, high thermal stability and wide electrochemical window.¹ These liquid salts, that are typically composed of organic cations and a variety of inorganic or organic anions, have various functional groups that are often tuneable to suit specific applications. This property is especially useful in designing ionic liquids as electrolytes for rechargeable batteries, such as those based on lithium (Li), sodium (Na) or magnesium (Mg), where the cations and anions can play a key role in the metal ion coordination and the electrochemical stability of the electrolyte.

The most successfully used ionic liquids for Li and Na battery applications typically incorporate ammonium, phosphonium or pyrrolidinium cations coupled with a fluorinated anion such as bis(fluorosulfonyl)imide ((FSO₂)₂N[−], FSI) or bis(trifluoromethanesulfonyl)imide ((CF₃SO₂)₂N, NTf₂).^{2–4} One of the key challenges in designing ionic liquids lies in the stability of the electrolyte itself or both the electrolyte/anode and the electrolyte/cathode interphases. For instance, an anion with insufficient oxidative stability can limit the choice of cathode material.^{5,6} Likewise, the

Broader context

In an effort to reduce the world's fossil fuel consumption, alternative cost-efficient, reliable and sustainable renewable energy storage devices need to be developed. High-energy density devices such as rechargeable lithium and magnesium batteries have gathered substantial interest in many energy storage applications such as electric vehicles (EV). However, the use of conventional electrolytes of high-flammability creates a safety issue in batteries as well as limiting the use of some low voltage anodic or high voltage cathodic materials. Therefore, the demonstration of non-flammable and thermally stable electrolytes with wide electrochemical windows are of significance for the next generation of high-energy density batteries.

instability of the anion in the highly reducing environment at the anode could lead to passivation, resulting in poor rechargeability. This becomes especially apparent for divalent magnesium or calcium electrochemistry due to the low reductive stability of anions typically used, such as [NTf₂][−].^{7–10} For example, it is known that ionic liquids are incompatible with Mg metal chemistries,^{11,12} where Mg deposition/stripping reversibility can only occur in the presence of highly reducing or corrosive chloride based Mg reagents/salts,^{13,14} often coupled with a volatile solvent like tetrahydrofuran.^{12,15–17} This, in addition to the low oxidative stability of these reagents (less than 2.5 V vs. Mg/Mg²⁺), does not represent an ideal situation.

The pursuit of rechargeable Mg batteries has been motivated by the high volumetric capacity of Mg (3832 mA h cm^{−3}), its negative reduction potential (−2.37 V vs. SHE) and absence of dendrite formation during deposition.^{7–9} While the lack of high-performing Mg salts with wide electrochemical windows previously plagued this technology, recent development of simple Mg salts based on boron clusters allowed this major challenge to be overcome.^{8,18–20} However, one key issue with all Mg electrolytes that has not been addressed so far is associated with the dependence on ethereal solvents, owing to their unique combination of solvation power and ability to withstand the electrochemical reductive environment of the magnesium anode.²¹ For several important reasons these solvents are not

^a ARC Centre of Excellence for Electromaterials Science, School of Chemistry, Monash University, Victoria 3800, Australia

^b Toyota Research Institute of North America, Materials Research Department, 1555 Woodridge Avenue, Ann Arbor, Michigan, USA. E-mail: rana.mohtadi@toyota.com

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ideal for optimum battery performance. Firstly, the anodic stability of these solvents is limited to about 3.3 V vs. Mg/Mg²⁺²² thereby challenging the development of high voltage cathodes for Mg batteries (3 V and above class). Secondly, the smaller ether solvents, such as tetrahydrofuran (THF) and dimethoxyethane (DME), display high volatility and produce electrolyte solutions that are not practical given the potential safety risks they impose. Thirdly, the strong binding between the magnesium ion and ether solvents^{23,24} results in high desolvation energy barriers that may lead to large overpotentials during battery operation.

Given the aforementioned challenges, ILs are potential candidates for replacing ether solvents in Mg batteries. Moreover, it would be ideal if these could be extended to other battery chemistries and have the potential to overcome existing challenges including those associated with limited anodic stability of the electrolyte. To enable this, we exploit in this work *closo*-boron cluster anions, as their Mg salts were shown to be highly compatible with Mg metal.^{18,20} Particularly, *closo*-monocarboranes are attractive in this regard, since they have a high degree of charge delocalisation over the entire cluster, making them weakly coordinating and relatively inert.^{8,25} Recent work has also shown that these anions have high anodic stability (*i.e.* 4.9 V vs. Mg/Mg²⁺), which is an indispensable property in the pursuit of high voltage battery chemistries.²⁰ However, the structural symmetry of these anions has been detrimental towards the formation of room temperature ionic liquids. In fact, there has been only a handful of *closo*-monocarborane based ionic liquids reported, such as those containing imidazolium^{26,27} and pyridinium²⁶ cations, albeit mostly having a high melting point. Amongst them, *N*-pentylpyridinium, [C₅pyr]⁺ [closo-CB₁₁H₁₂][−] at 19 °C²⁸ displays the lowest melting point reported in the literature, even though this value is slightly surprising in light of reported melting points of 134 °C and 114 °C for the closely related [C₄pyr]⁺ and [C₆pyr]⁺ salts, respectively.²⁶ All other known salts have a much higher melting point. For example, the imidazolium family salts have melting points of 70 °C for [C₈mim]⁺²⁶ and of 122 °C for [C₂mim]⁺.²⁷

The reported melting points for ammonium-based salts were all above 100 °C.²⁹ These very high melting points have made ionic liquids based on *closo*-boron clusters unsuitable for ambient temperature battery applications including electric vehicles (EV), which typically require electrolytes to be liquids down to −30 °C. In the present work, we hypothesized that the structural rigidity of the cluster, which tends to promote elevated melting points, could be compensated for by careful design of the cation. Thus, we demonstrate a proof of concept that ionic liquids with very low melting temperatures can be prepared, for the first time, based on the symmetrical and highly stable *closo*-monocarborane anion [CB₁₁H₁₂][−]. These ionic liquids possess excellent physical and electrochemical properties and compatibility with reactive metals, demonstrating that ILs based on *closo*-boron clusters are promising new candidates for energy storage systems. A comparison of some key properties of various ionic liquid battery electrolytes is summarised in Table 1.

Two ionic liquids were designed and synthesised based on an alkoxy ammonium cation and the [CB₁₁H₁₂][−] anion (Fig. 1).[‡] This cation was selected owing to its high reductive stability

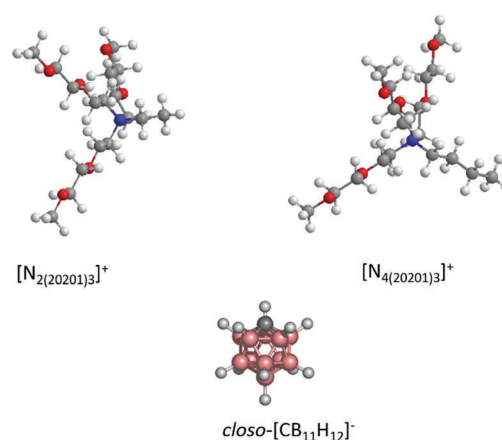


Fig. 1 Structures of [N₂₍₂₀₂₀₁₎₃][CB₁₁H₁₂], **IL1**, and [N₄₍₂₀₂₀₁₎₃][CB₁₁H₁₂], **IL2**. (N = blue, O = red, C = grey, H = white, B = pink.)

Table 1 Comparison of melting point, glass transition, anodic stability and comments on Mg compatibility/properties of [N₄₍₂₀₂₀₁₎₃][closo-CB₁₁H₁₂] with previously reported ionic liquids

Ionic liquid	<i>T</i> _m (°C)	<i>T</i> _g (°C)	Anodic stability (V vs. Fc/Fc ⁺)	Comments
[C ₃ mpip][NTf ₂] ³⁰	8.7	—	2.2	– Not compatible with Mg – Too high <i>T</i> _m
[C ₅ pyr][closo-CB ₁₁ H ₁₂] ³¹	19	—	NA	– Cation not reductively stable – Too high <i>T</i> _m
[C ₈ mim][closo-CB ₁₁ H ₁₂] ²⁶	70	—	NA	– Cation not reductively stable due to C2 proton on cation – Too high <i>T</i> _m
[N ₂₍₂₀₂₀₁₎₃][NTf ₂] ³² [MPEG _n][NTf ₂] ¹⁴ (<i>n</i> = 3, 7)	—	−68	1.6* −1.37**	*Not compatible with Mg **Only cycled Mg when Mg[BH ₄] ₂ was used, [BH ₄] [−] oxidative stability is −1.37 V vs. Fc/Fc ⁺
This work [N ₄₍₂₀₂₀₁₎₃][closo-CB ₁₁ H ₁₂]	—	−52	1.7	– Compatible with Mg (& Li) – High reductive stability

Note: for convenience we have converted and reported the anodic stability all ILs vs. Fc/Fc⁺.

together with moderately long and flexible side groups that typically induce low melting point through the introduction of disorder and asymmetry.^{13,33,34} The incorporation of an ether motif into the ammonium cation is utilised here to dominate interactions with the anion, as shown in previous work,¹³ thereby assisting with solubility of metal salts in the IL. An increase in alkyl chain length from ethyl to butyl in the cation $[N_{x(20201)3}]^+$ ($x = 2$ in **IL1** or 4 in **IL2**) is also investigated here since the larger ion typically leads to lower melting point.³⁵ Full details of the preparation and characterisation of the ILs can be found in the ESI.†

The as-synthesised **IL1** and **IL2** are observed to be in a molten state at room temperature and the thermal properties shown in Fig. 2a clearly support that both **IL1** and **IL2** are liquids below room temperature, down to -52°C , there being no melt or freezing transitions in the DSC traces. The observed glass transition temperature (T_g) is by far the lowest reported to date for any *closo*-boron cluster type ionic liquids (by *ca.* 70°C) and notably even comparable to those widely used ILs based on $[\text{NTf}_2]^-$ anion. For example, compared to **IL1**, ($T_g = -52^\circ\text{C}$), the analogous $[\text{NTf}_2]^-$ salt has previously been shown to exhibit a T_g around -68°C .³² These results clearly demonstrate that the role of the cation is key in designing ILs with low melting point, where the ether functionalised alkyl chains effectively disrupt the crystal packing.²⁷ The lower T_g in $[\text{N}_{2(20201)3}][\text{NTf}_2]$ can be attributed to the greater conformational freedom of $[\text{NTf}_2]^-$ anion³⁶ when compared to the rigid structure of $[\text{CB}_{11}\text{H}_{12}]^-$. Nonetheless, the shear ability to achieve such low glass transition point using rigid and symmetrical carborane anion that is

only 16°C higher than that of the flexible $[\text{NTf}_2]^-$ based ILs is fascinating. When comparing **IL1** and **IL2**, a small increase in T_g (5°C) is observed in **IL2** when the chain length is increased from ethyl to butyl. This suggests that further increase in the alkyl chain length in these cations does not necessarily lower the T_g , as usually observed with other ionic liquids,³⁷ likely due to the already lengthy ether moieties.

Another important property of ionic liquids is their thermal stability which determines their operational thermal window. In the thermal gravimetric analysis (TGA) experiments conducted at $10^\circ\text{C min}^{-1}$, the decomposition for both **IL1** and **IL2** does not begin until approximately 300°C , as shown in Fig. 2b. A rapid mass decrease is then observed up to 350°C . This high thermal stability is similar to other ILs with weakly coordinating anions such as $[\text{NTf}_2]^-$, which tend to have higher decomposition temperatures than other RTILs (*i.e.* decomposition for $[\text{N}_{2(20201)3}][\text{NTf}_2]$ is 300°C).^{38,39} Note that the high stability is consistent with that of the inorganic salts incorporating the same anion, which is typical of these boron clusters.²⁰ The similar decomposition temperature of **IL1** and **IL2** suggests that decomposition of both these ionic liquids occur in a similar manner *via* the reverse Menshutkin/Hoffman elimination mechanism where the anion may be attacking the C2 in the ether groups of the ammonium cation and not the alkyl chains.⁴⁰

Motivated by these promising results, we investigated the transport and dissociative properties of these ionic liquids. The viscosity and ionic conductivity are shown in Fig. 3a and b, respectively. The two ILs have similar transport properties; for example, at 30°C , the viscosity of **IL1** is $650 \pm 3 \text{ mPa s}$ while the viscosity of **IL2** is $652 \pm 3 \text{ mPa s}$. Similarly, the ionic conductivity at 30°C of **IL1** ($1.1 \pm 0.1 \times 10^{-4} \text{ S cm}^{-1}$) and **IL2** ($1.02 \pm 0.1 \times 10^{-4} \text{ S cm}^{-1}$) are very close. Their room temperature conductivities are more than an order of magnitude higher than those of reported ILs based on $[\text{C}_2\text{mim}]^+$ cation with the structurally similar *closo-o*-carboranyl anion $[\text{closo-1,2-C}_2\text{B}_{10}\text{H}_{11}]^-$ ($2.9 \times 10^{-5} \text{ S cm}^{-1}$) at 51°C .⁴¹ Besides the possible contribution of anion charge localization, such difference suggests the role of the ether functionalized ammonium cation in making **IL1** and **IL2** more fluid. The dissociative behaviour of ionic liquids is typically judged through constructing the Walden plot, which describes the deviation from a highly dissociative reference (aqueous KCl). The plot in Fig. 3c, which demonstrates a relationship between the molar conductivity and the fluidity of the ILs ($A = \eta^{-1}$),^{42,43} shows that both ionic liquids consist of relatively independently mobile ions, consistent with desired high ionicity. In fact, the ionicity of the carborane ionic liquids are comparable to other ether-based ionic liquids such as $[\text{N}_{2(20201)3}][\text{NTf}_2]$ ³² as well as conventional ionic liquids $[\text{C}_4\text{mpip}][\text{NTf}_2]$.⁴⁵ In addition, these ionic liquids sit considerably closer to the ideal line in the Walden plot than many viscous ILs such as $[\text{P}_{66614}][\text{NTf}_2]$ ^{43,44} suggesting that $[\text{CB}_{11}\text{H}_{12}]^-$ is exhibiting the highly desired characteristics of a weakly coordinating anion in these ILs.

Having established, for the first time, that ionic liquids based on a *closo*-monocarborane anion could possess properties that make them relevant for energy storage applications,

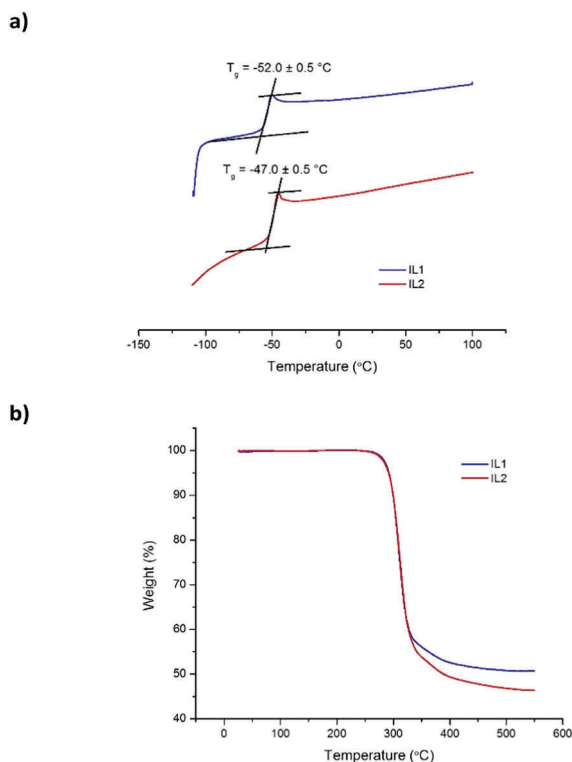


Fig. 2 Thermal properties of **IL1** and **IL2**; (a) DSC trace and (b) TGA.

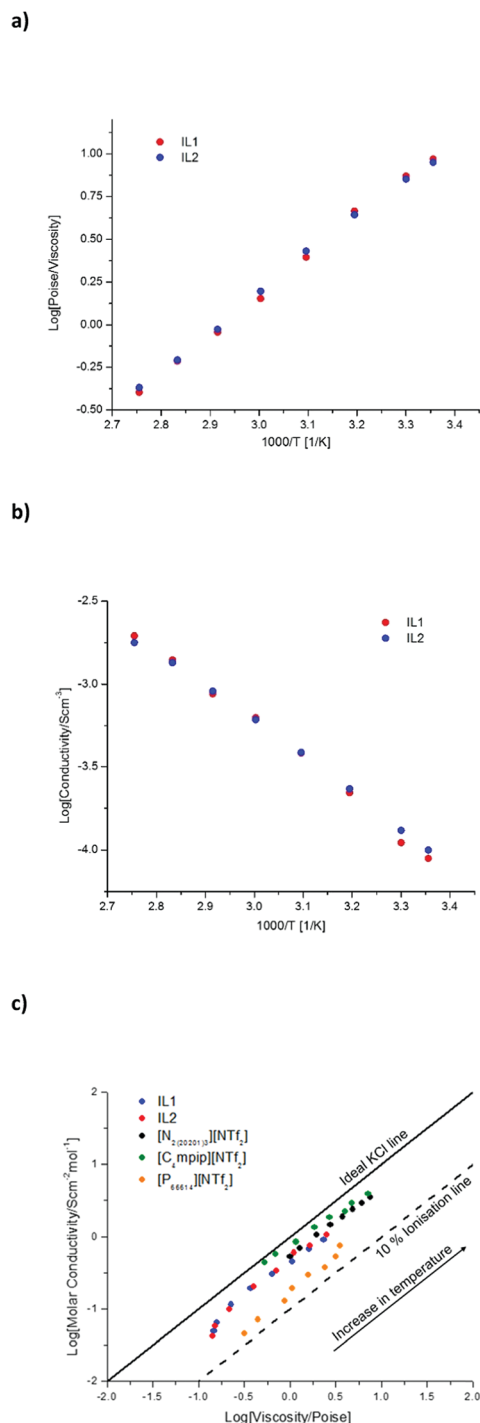


Fig. 3 (a) Viscosity (b) conductivity and (c) Walden plot (ionicity data) for **IL1** and **IL2** in comparison with ionic liquids from previous reports.^{32,44,45} Solid line represents the ideal KCl line. Dashed line represents molar conductivity which is 10% of the ideal KCl case.

their electrochemical oxidative and reductive stabilities were studied. Fig. 4 illustrates the linear sweep voltammograms (LSVs) of **IL1** and **IL2**. As we hypothesized, the cathodic stability is found to be high, down to about -3.3 V vs. Ag/Ag^+ (*ca.* -0.05 V vs. Li/Li^+) for **IL2**. Such a value allows for utilization of these ionic liquids for battery chemistries with low operating

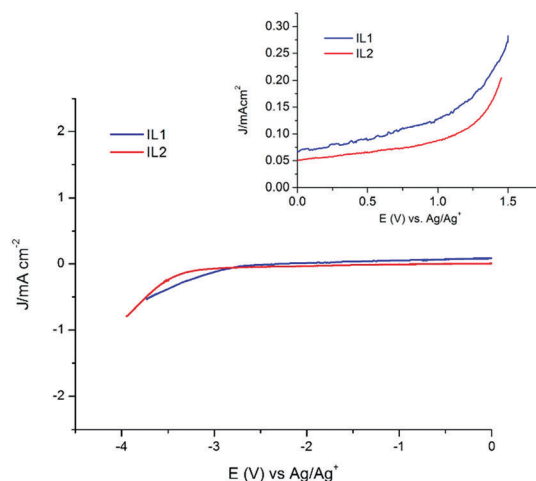


Fig. 4 Linear sweep voltammograms of **IL1** and **IL2** vs. Ag/Ag^+ (working electrode – GC, 25 °C) depicting the reductive scan. The inset shows the oxidative scan.

potentials such as those based on Mg and Li metals. The anodic stability limit is also high at about $+1.25$ V vs. Ag/Ag^+ (*ca.* 4.5 vs. Li/Li^+), approximately 0.5 V above that of ethereal solvents and consistent with that shown by Tutusaus *et al.* for the $\text{Mg}(\text{CB}_{11}\text{H}_{12})_2$ salt (anodic stability of *ca.* 5.6 vs. Li/Li^+).²⁰ As the alkoxy chains in the ammonium cation dictate this stability, we are confident that future designs that don't incorporate these alkoxy moieties would even result in a higher anodic stability approaching that of the $[\text{CB}_{11}\text{H}_{12}]^-$ anion.

The compatibility of the ILs with Mg metal was examined using $\text{Mg}[\text{BH}_4]_2$ and $\text{Mg}[\text{CB}_{11}\text{H}_{12})_2]$ (MMC) salts (see ESI† for preparative details). The solubility of $\text{Mg}[\text{BH}_4]_2$ was close to 0.2 M and both Mg reduction and oxidation was observed suggesting the compatibility of these ILs with Mg metal (see ESI† Fig. S3b). However, the hydridic nature of the borohydride that makes it highly reducing was shown to mitigate the effects of Mg passivation even by the $[\text{NTf}_2]^-$ anion.¹³ Therefore, to get a conclusive support on the suitability of the IL for Mg metal electrochemistry, the use of an inert Mg compatible salt is required and hence MMC salt is an ideal choice. The MMC initially displays very low solubility (<0.01 M in **IL2**, Fig. 5), likely caused by inability with of the alkoxy chain in the

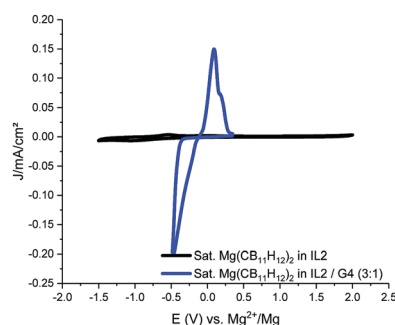


Fig. 5 CV on Pt electrode of a saturated $\text{Mg}(\text{CB}_{11}\text{H}_{12})_2$ solution in **IL2** (black) (cycle 10) and in **IL2**/G4 (3:1) (cycle 10).

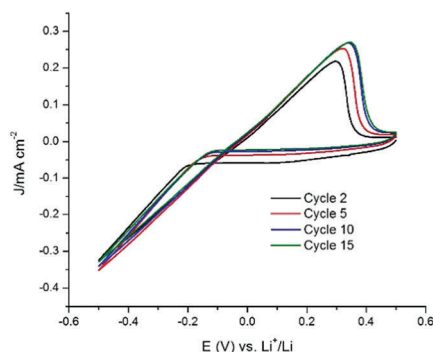


Fig. 6 CV of 0.2 M Li[NTf₂] in IL2 over 15 cycles under inert conditions. (RE: Li, CE: Li, WE: Ni) at 60 °C.

ammonium cation to exchange the strongly coordinating solvent (DME) attached to the Mg²⁺ in the as-prepared MMC salt. Therefore, to allow for interrogating the compatibility of IL2 with Mg, we found that adding a small amount of tetraglyme (G4) to IL2 results in enhanced solubility of the salt (in the long run, using an ether-free MMC salt is desired and calls for development of new synthetic strategies capable of producing this salt). Gratifying, as shown in Fig. 5, the CV results reveal compatibility of the IL with Mg metal, as evidenced in the coulombic efficiency of 70% being sustained over 50 cycles (see ESI,† Fig. S4) and typical smooth morphologies of deposited Mg (Fig. S5–S10, ESI†). Given that no Mg cyclability has been thus far possible in other ILs, these carborane ionic liquids show great potential as alternatives to ether solvents that support reversible Mg electrochemistry.

The compatibility of these ionic liquids with Li metal was also investigated. Li cycling has previously been demonstrated in many ionic liquids commonly composed of pyrrolidinium^{46–50} and recently phosphonium² cations with fluorinated anions.^{50,51} Here the electrochemical reduction and oxidation of Li from 0.2 M Li[NTf₂] in IL2 at 60 °C is presented in the CV of Fig. 6. The *closo*-monocarborane clearly demonstrates the ease of Li reduction and oxidation without any activation. This also highlights the thermal stability of these electrolytes. The Li cyclability is in line with that reported for the LiCB₁₁H₁₂ solid state electrolyte²⁵ and demonstrates that *closo*-monocarborane based ionic liquids represent new potential candidates as electrolytes for lithium batteries. This unprecedented compatibility with both Li and Mg metal, despite the different requirements (*i.e.* no SEI for Mg metal), makes ILs based on *closo*-boron cluster anions the first of a kind and highlights their unique versatility.

In conclusion, we unveiled our design strategy that enables the formation of *closo*-boron cluster type ionic liquids which are relevant to energy storage devices. The first examples of very low melting temperature *closo*-monocarborane based RTILs have been reported here, where both [N₂(20201)₃][CB₁₁H₁₂] and [N₄(20201)₃][CB₁₁H₁₂] display unprecedented low *T_g* at approximately −52 °C and −47 °C, respectively. Also, we demonstrate for the first time Mg compatible ionic liquids. This, coupled with compatibility with Li metal and a wide electrochemical window (4.55 V), make ionic liquids based on the

closo-monocarborane anion serious contenders as solvents for various rechargeable metal batteries. This proof of concept represents the beginning of a new design space of ILs that exploits the properties and rich chemistry of boron clusters to create highly performing ionic liquids for energy storage devices.

Conflicts of interest

There are no conflicts to declare.

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‡ Experimental section: The synthesis and characterisation of both *N*-ethyl-*N,N,N*-tris(2-(2-methoxyethoxy)ethyl)ammonium 1-carba-*closo*-dodecaborate [N₂(20201)₃][CB₁₁H₁₂] (IL1) and *N*-butyl-*N,N,N*-tris(2-(2-methoxyethoxy)ethyl)ammonium 1-carba-*closo*-dodecaborate [N₄(20201)₃][CB₁₁H₁₂] (IL2), is demonstrated in the ESI.† Magnesium borohydride (Mg[BH₄]₂, 95%) and lithium bis(trifluoromethylsulfonyl)imide (LiNTf₂) were purchased from Sigma-Aldrich. Caesium 1-carba-*closo*-dodecaborate (CsCB₁₁H₁₂) was purchased from Katchem and magnesium 1-carba-*closo*-dodecaborate, Mg[CB₁₁H₁₂]₂ (MMC) was synthesised according to literature.²⁰ Full experimental details are provided in ESI.†

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